

IoT Based Smart Robotic Assistance for Alzheimer's Patient

Presented by

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Introduction, Problem Statement and Aim

1. An AI-based system that monitors patients using computer vision.
2. Alzheimer Patients face free fall and memory lose, needs continuous assistance, unfortunately difficult for caregivers.
3. Aim is to design mobility robot for monitoring and following patient ensuring continuous assistance.





Modules

- Fall Detection
  Conversation of robot with patient
- Patient tracking and following
- Environment Monitoring + Robotic Model
- App Integration

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Implementation of Dataset Creation(1/2)

Phase 01

Phrase 01:

Tables 1 Showing the phase 01 of dataset creation.

Classes	Total videos	Place 1 Left angle Member 1	Place 2 Right angle Member 2	Place 3 Front angle Member 3	Place 4 Back angle Member 4
Falling	40	10	10	10	10
Sitting	40	10	10	10	10
Walking	40	10	10	10	10
Sleeping	40	10	10	10	10
Total	**160**				

phase 02

Phrase 02:

Tables 2: Showing the phase 02 of dataset creation.

Classes	Total videos	Place 1 Left angle Member 5	Place 2 Right angle Member 5	Place 3 Front angle Member 6	Place 4 Back angle Member 6
Falling	80	20	20	20	20
Sitting	80	20	20	20	20
Walking	80	20	20	20	20
Sleeping	80	20	20	20	20
Total	**320**				

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Implementation of Dataset Creation(2/2)

Dataset Creation Phase Completed.

Tables 3: Showing the data in each class of final dataset.

Classes	Total no of videos
Falling	120
Sitting	120
Walking	120
Sleeping	120
Total	480

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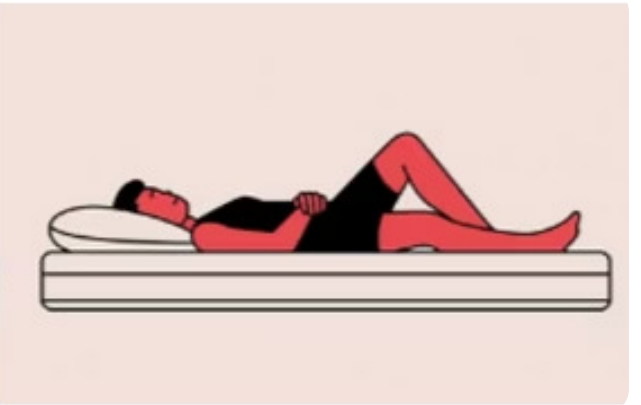
Limitations in fall Detection by Basic Python

- Falling and sleeping posture is same.
- In order to make difference we need temporal analysis along with spatial analysis.









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Methodology of Model Training

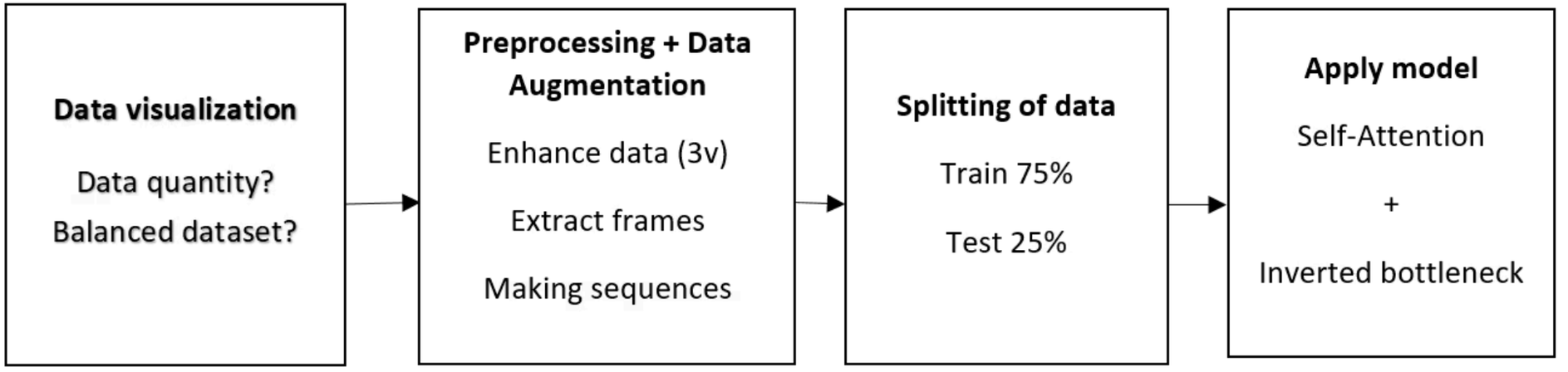


Figure: Pipeline for training the model

Tables 4: Distribution of data keeping FPV=20, SL=20.

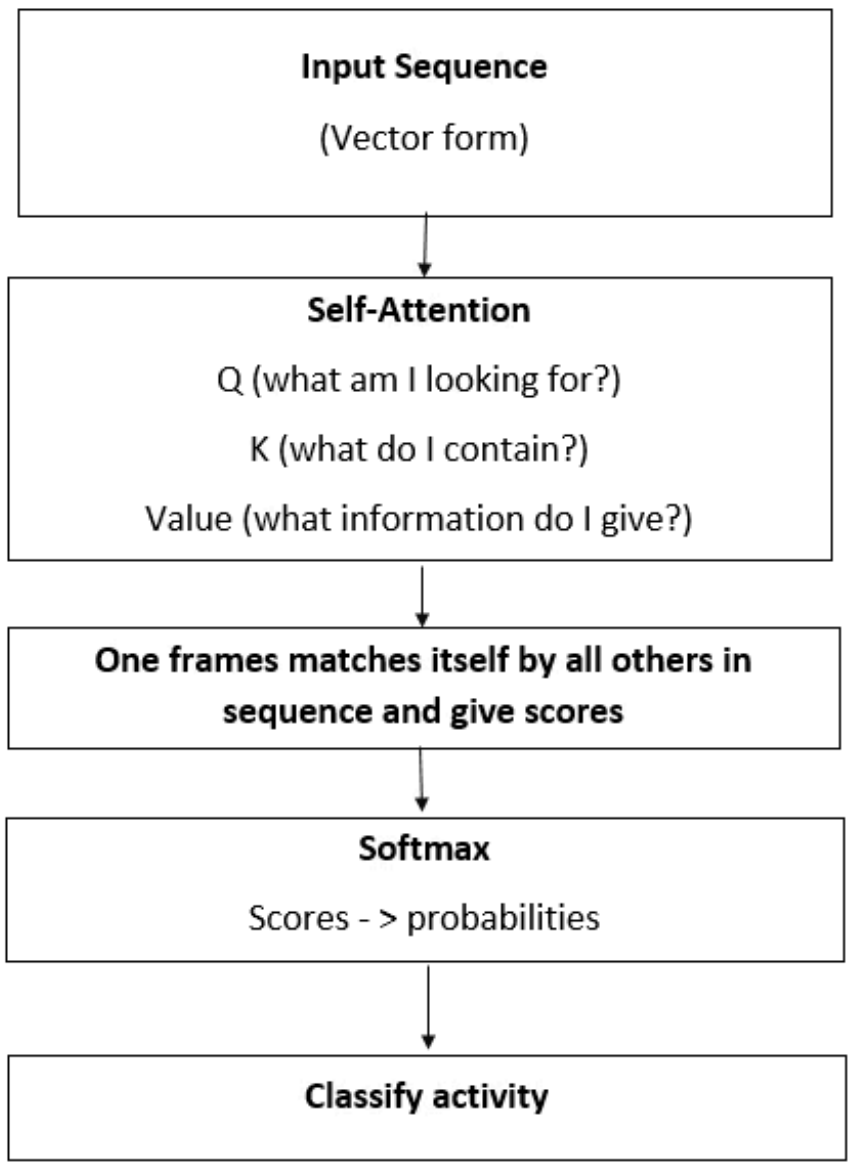
Classes	Total videos before augmentation	Total videos after augmentation	Sequences with SL = 20	Train data	Test data
Falling	120	120*4 = 480	480	360	120
Sitting	120	120*4 = 480	480	360	120
Sleeping	120	120*4 = 480	480	360	120
Walking	120	120*4 = 480	480	360	120
Total	**480**	**1920**	**1920**	**1440 (75% of 1920)**	**480 (25% of 1920)**

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Algorithm and Flow Diagrams of Models

1

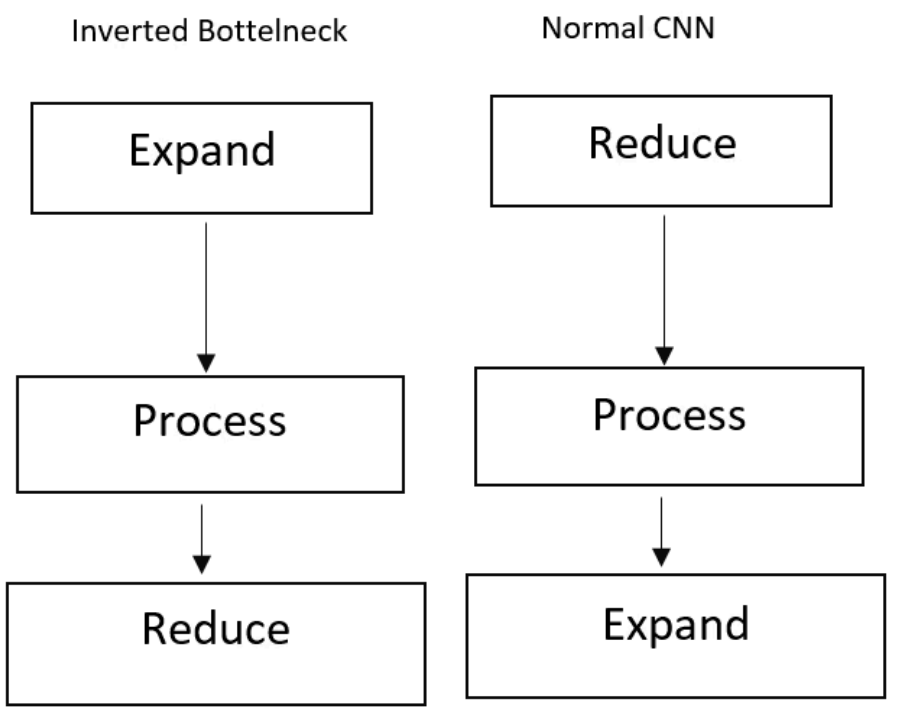
Algorithm for Self-Attention Model



2

Flow diagram of InvertedBottleneck Model

- Efficient features extraction
- With Fewer Parameters
- For better performance



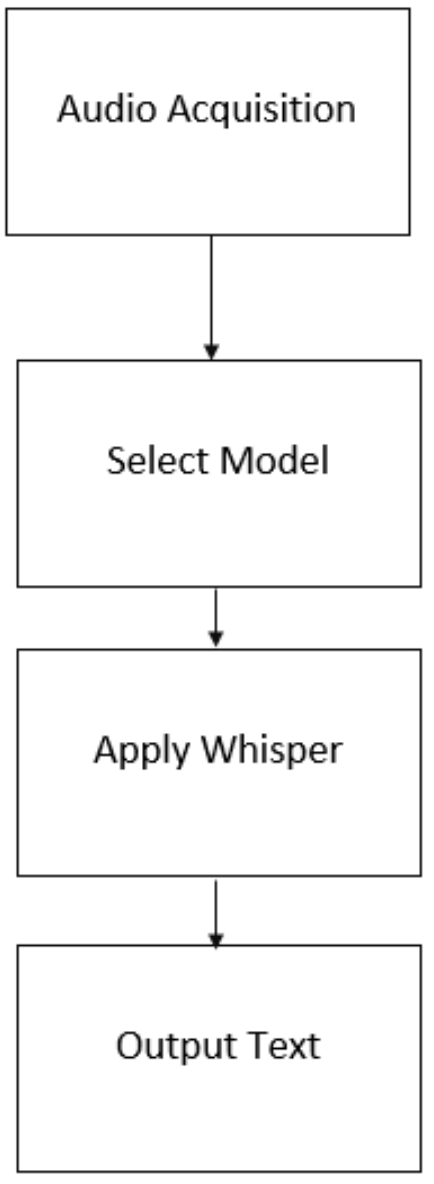
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Implementation of STT and TTS Conversion

1

Methodology for SST

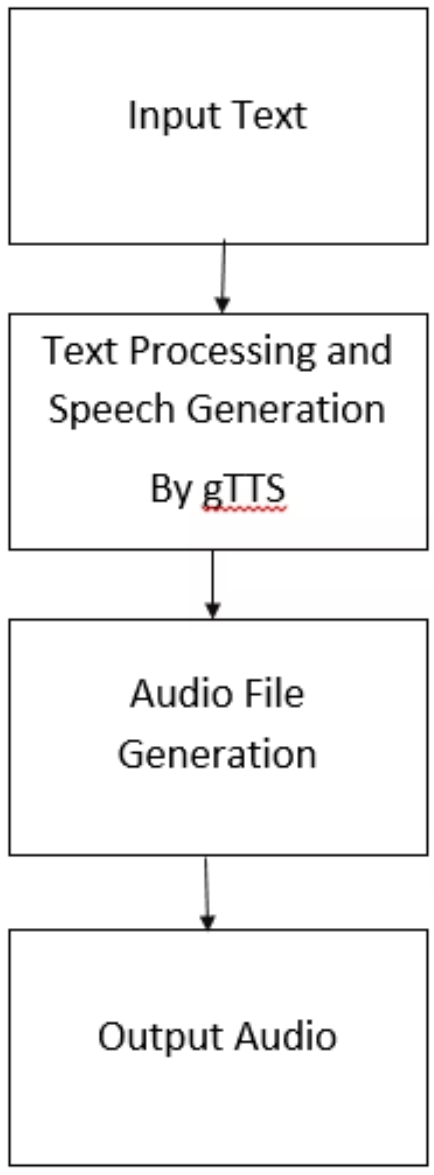
- Patient voice —› Robot
- By whisper deep learning model.



2

Methodology for TTS

- Robot Voice —› Patient
- By using gTTS Model.



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Implementation of Patient Tracking

Methodology of Patient Tracking and Following

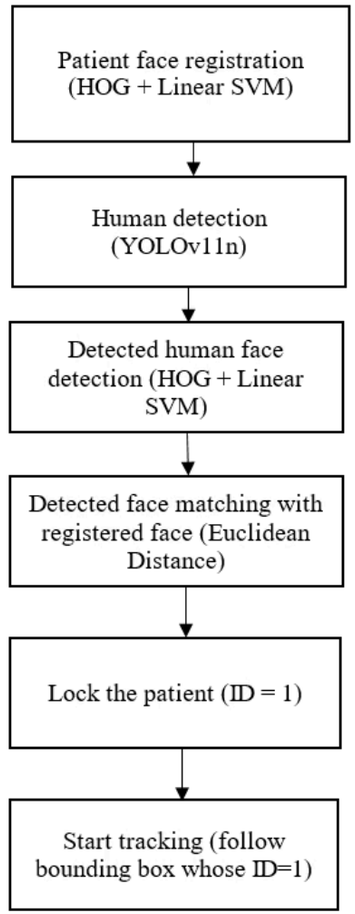


Figure: Pipeline for Patient tracking

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Data Collection and Wifi Connection



Data Readings & WiFi Connection ESP32

- ESP32 is connected to sensors:
- Reads analog/digital data from sensors
- ESP32 Connects to WiFi network
- Arduino IDE monitor shows results

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Sending data to Firebase Realtime Database



Data ESP32 to Firebase

- ESP32 sends processed data to Firebase using JSON
- Authenticates using Firebase API key
- Updates database nodes in real time
- ESP32 logs displayed on Firebase

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Real-time Data on Android App

Real-Time Data on Android App

- Android app connects to Firebase Realtime Database
- APP reads data of Firebase
- Displays real-time data on APP
  - Numeric values
- Shows Alerts
- Sent notification to Caregiver

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Pipeline for App integration with Hardware

Pipeline

- Sensors collect data → ESP32
- ESP32 processes & sends data → Firebase
- Firebase stores & updates data → Android App
- Android App displays data

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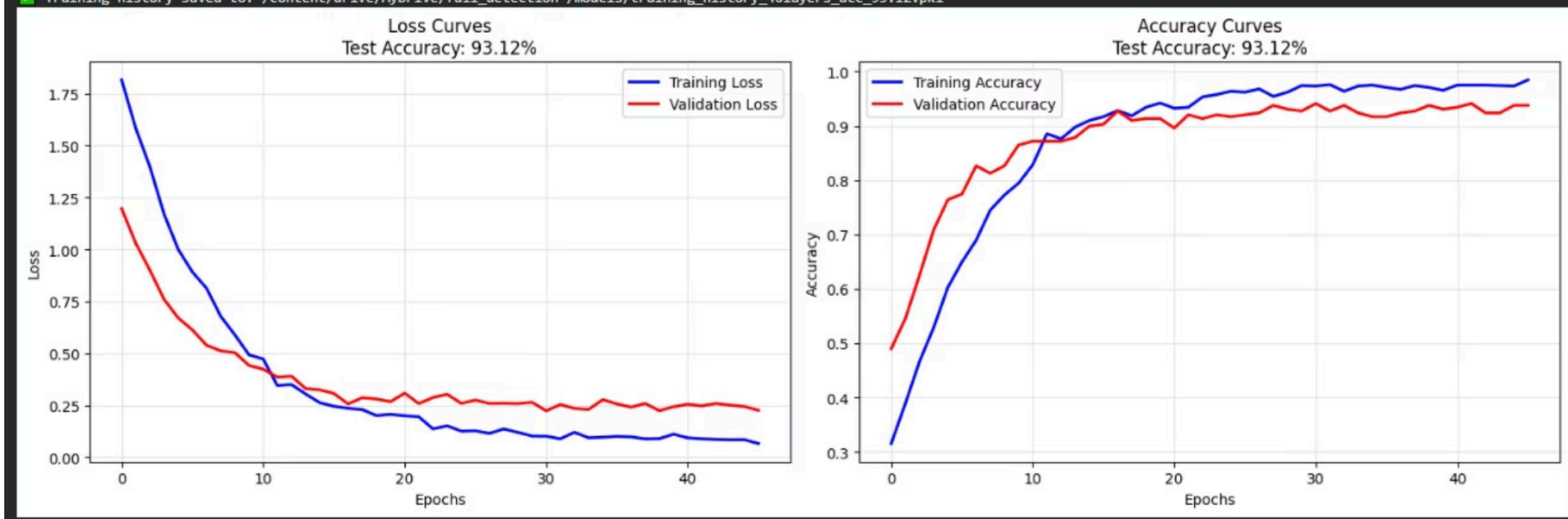
Results of Model Training

```
✅ Training completed!

============================================================
📊 EVALUATING MODEL ON TEST SET
============================================================
15/15 ━━━━━━━━━━━━━━━━━━━━ 52s 2s/step - accuracy: 0.9312 - loss: 0.2164
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file format is considered legacy. We recommend using instead the native Keras forma

🎯 Test Accuracy: 93.12%
📉 Test Loss: 0.2164

✅ Model saved to: /content/drive/MyDrive/fall_detection /models/inverted_bottleneck_40layers_attention_acc_93.12.h5
✅ Training history saved to: /content/drive/MyDrive/fall_detection /models/training_history_40layers_acc_93.12.pkl
```




```
✅ Training curves saved as 'inverted_bottleneck_40layers_training_curves.png'
```

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Results for SST and TTS Conversation

1

Result of SST

```
--------------------------------------------------

Enter your choice (1 or 2): 2

======================================================================
🎙️ LIVE RECORDING MODE
======================================================================

Enter recording duration in seconds (default: 5): 5

🔴 Recording will start in 2 seconds...
   Recording duration: 5 seconds
   Please speak clearly into your microphone
--------------------------------------------------
🔴 RECORDING...
✅ Recording complete! (Duration: 6.3 seconds)
💾 Audio saved to: recording_1774785439.wav

▶️ Playing back your recording...
```

0:04 / 0:04

```
======================================================================
🎙️ WHISPER TRANSCRIPTION
======================================================================

🚀 Loading Whisper model...
   Model options: tiny, base, small, medium, large
--------------------------------------------------

Choose model size (tiny/base/small/medium/large) [default: base]: base

🚀 Loading Whisper 'base' model...
✅ Model loaded in 1.45 seconds

🎤 Transcribing audio...
--------------------------------------------------
/usr/local/lib/python3.12/dist-packages/whisper/transcribe.py:132: UserWarning: FP16 is not supported on
  warnings.warn("FP16 is not supported on CPU; using FP32 instead")

======================================================================
📝 TRANSCRIPTION RESULT
======================================================================

🎙️ Recognized Text:
--------------------------------------------------
 Yes, I am fine.
--------------------------------------------------

📊 Details:
   Audio file: recording_1774785439.wav
   Model used: base
   Processing time: 4.18 seconds
```

2

Result of TTS

```
    return "output.mp3"

# Example
text_to_speech("heyy did you fall. are you okay. do you need help")
```

```
Collecting gtts
  Downloading gTTS-2.5.4-py3-none-any.whl.metadata (4.1 kB)
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.12/dist-packages (from
Collecting click<8.2,>=7.1 (from gtts)
  Downloading click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from reque
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from
Downloading gTTS-2.5.4-py3-none-any.whl (29 kB)
Downloading click-8.1.8-py3-none-any.whl (98 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 98.2/98.2 kB 4.9 MB/s eta 0:00:00
Installing collected packages: click, gtts
  Attempting uninstall: click
    Found existing installation: click 8.3.1
    Uninstalling click-8.3.1:
      Successfully uninstalled click-8.3.1
ERROR: pip's dependency resolver does not currently take into account all the packages that are in
typer 0.24.1 requires click>=8.2.1, but you have click 8.1.8 which is incompatible.
Successfully installed click-8.1.8 gtts-2.5.4
```

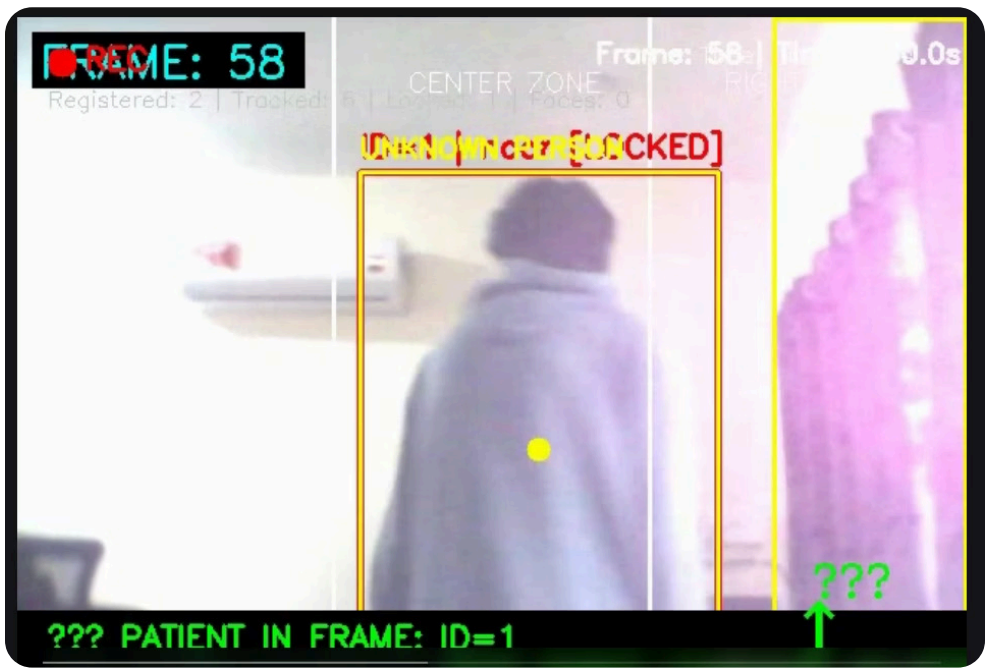
0:00 / 0:03

```
'output.mp3'
```

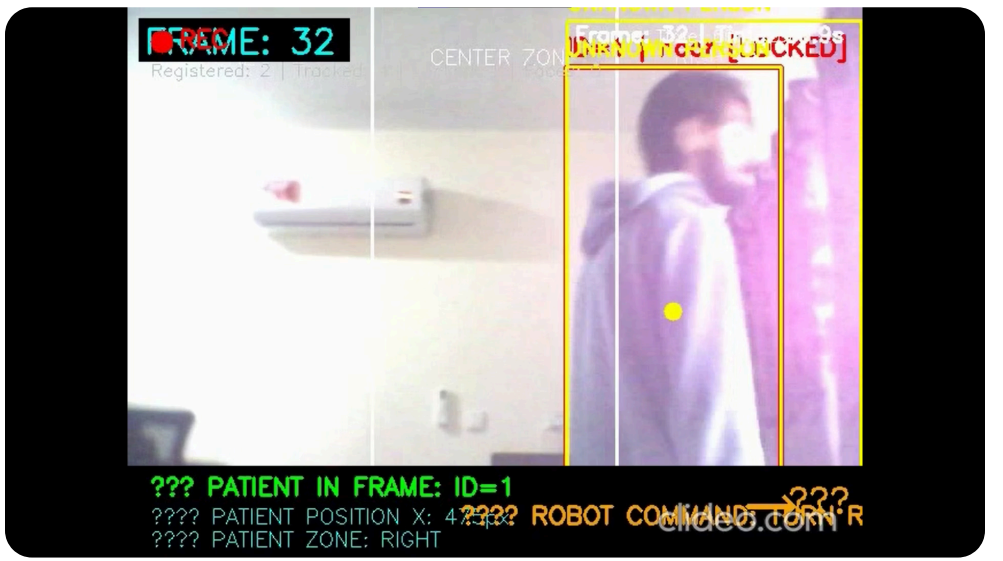
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Results for Patient Tracking

1



2

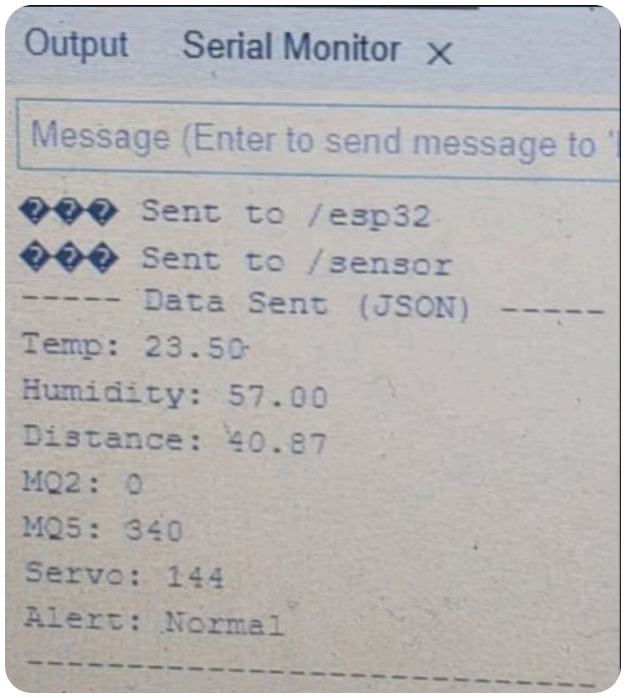


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Results of Sensors on Serial Monitor, Firebase and App

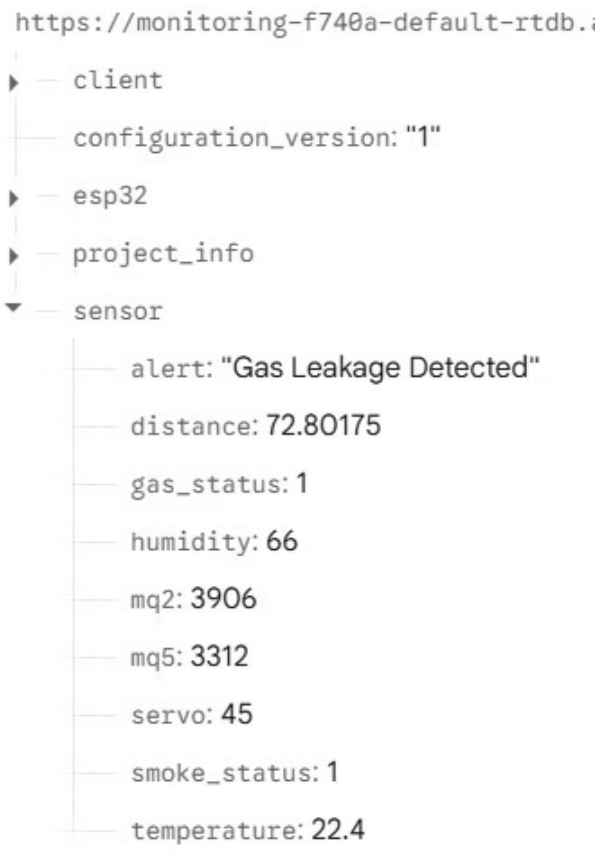
1

Readings on Serial Monitor



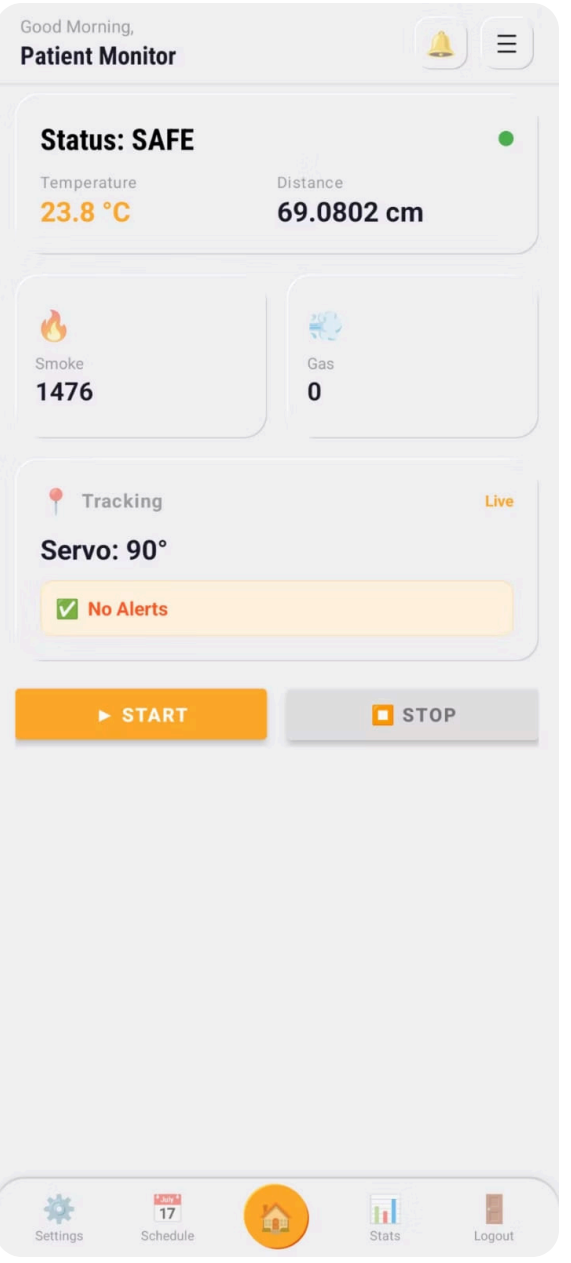
2

Readings on Firebase



3

Readings in App

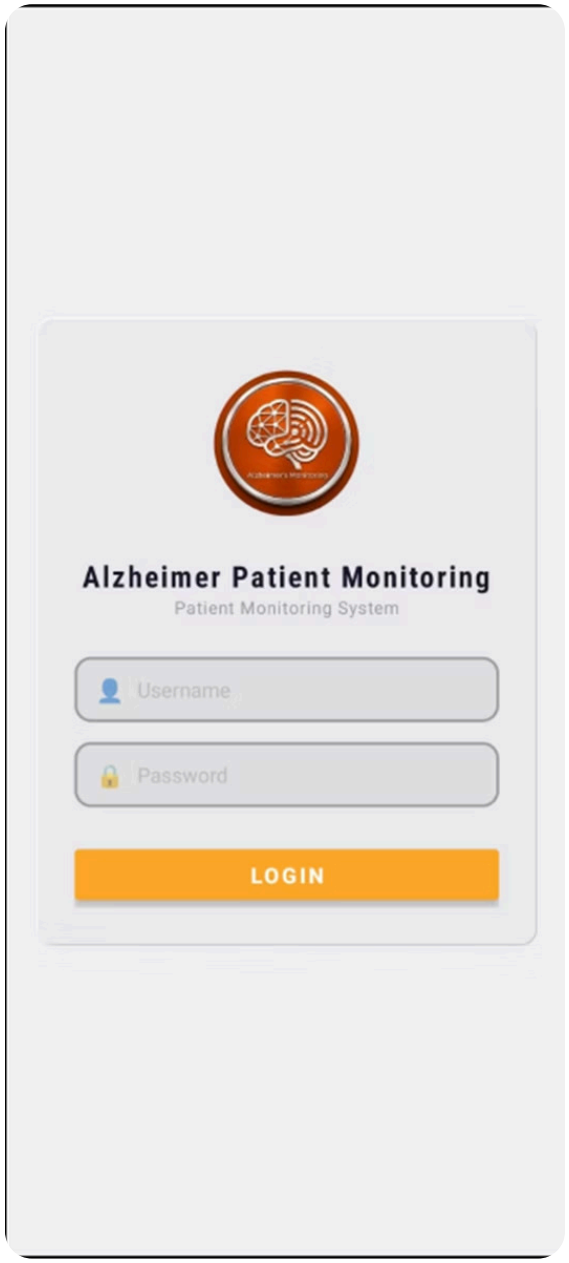


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Results for UI of App and Data Presentations

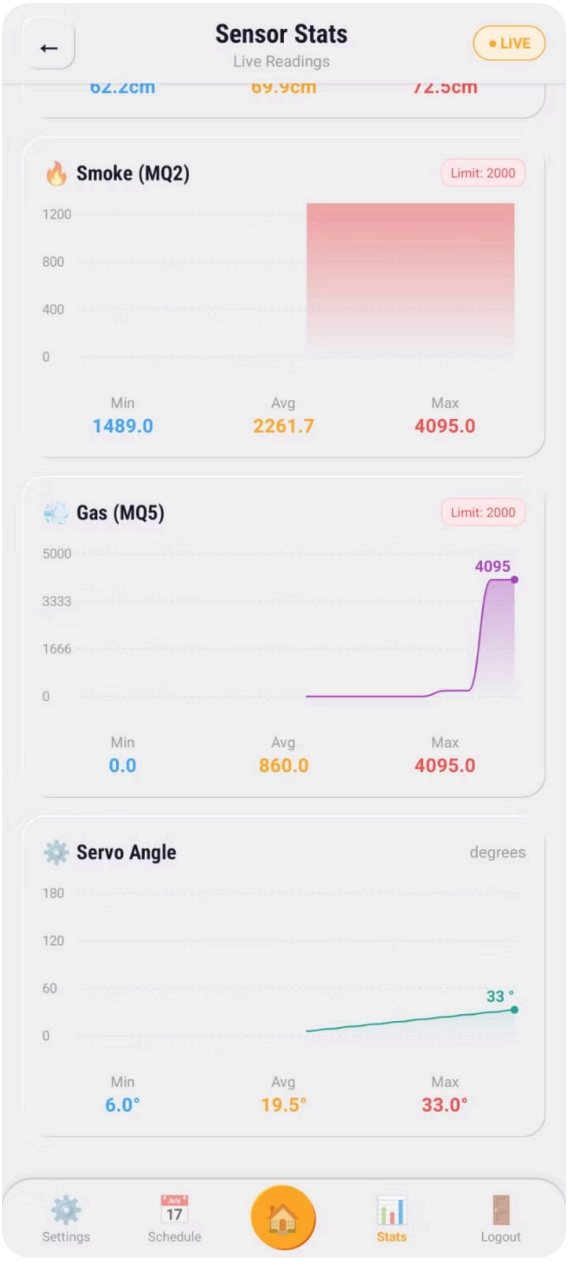
1

Login page



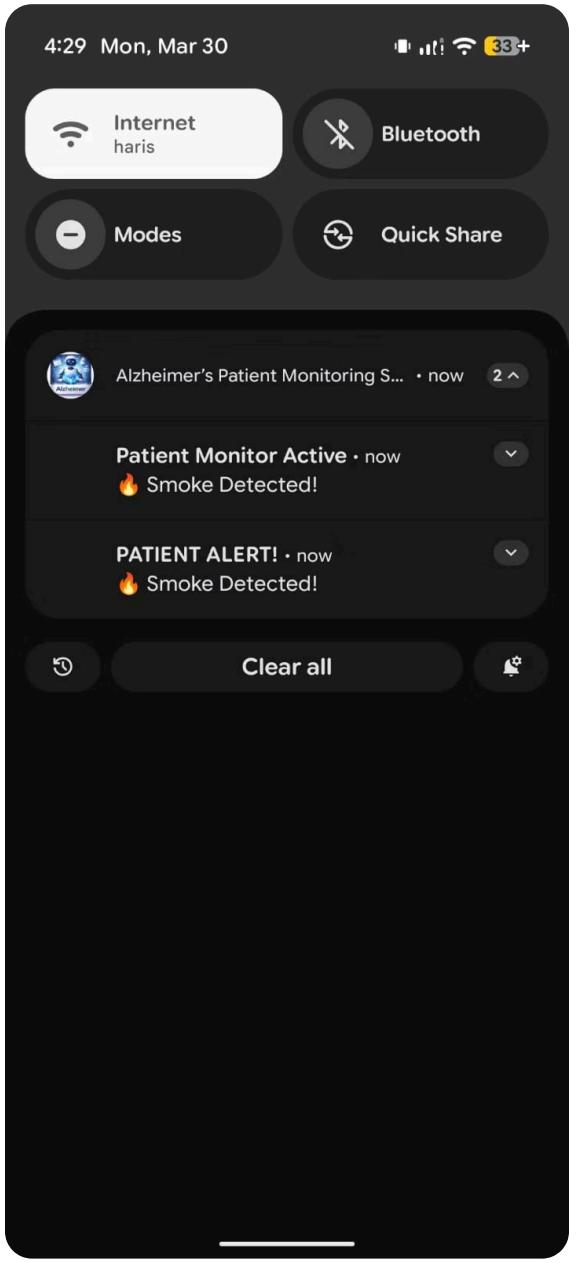
2

Graphical Representation of Readings



3

Generating Alerts



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Distribution of work



Noor uz Zaman

1. Customized fall dataset creation.
2. Implementation of HW on veroboard
3. Integration of Firebase with App



Zeenat Fatima

1. Model training for fall detection
2. Patient tracking and following.
3. Conversation between Patient and Robot.
4. Jetson OS installation



M Haris Ikram

1. Customized fall dataset creation.
2. Implementation of HW on veroboard.
3. Integration of Firebase with App
4. App development.

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TeamWork

Noor uz Zaman

1. Helped Zeenat in recording datasets
2. Implementation of HW on veroboard with Haris
3. Helped Haris in APP development

Zeenat Fatima

1. As a leader guided the workflow
2. Helped in distribution of work

M Haris Ikram

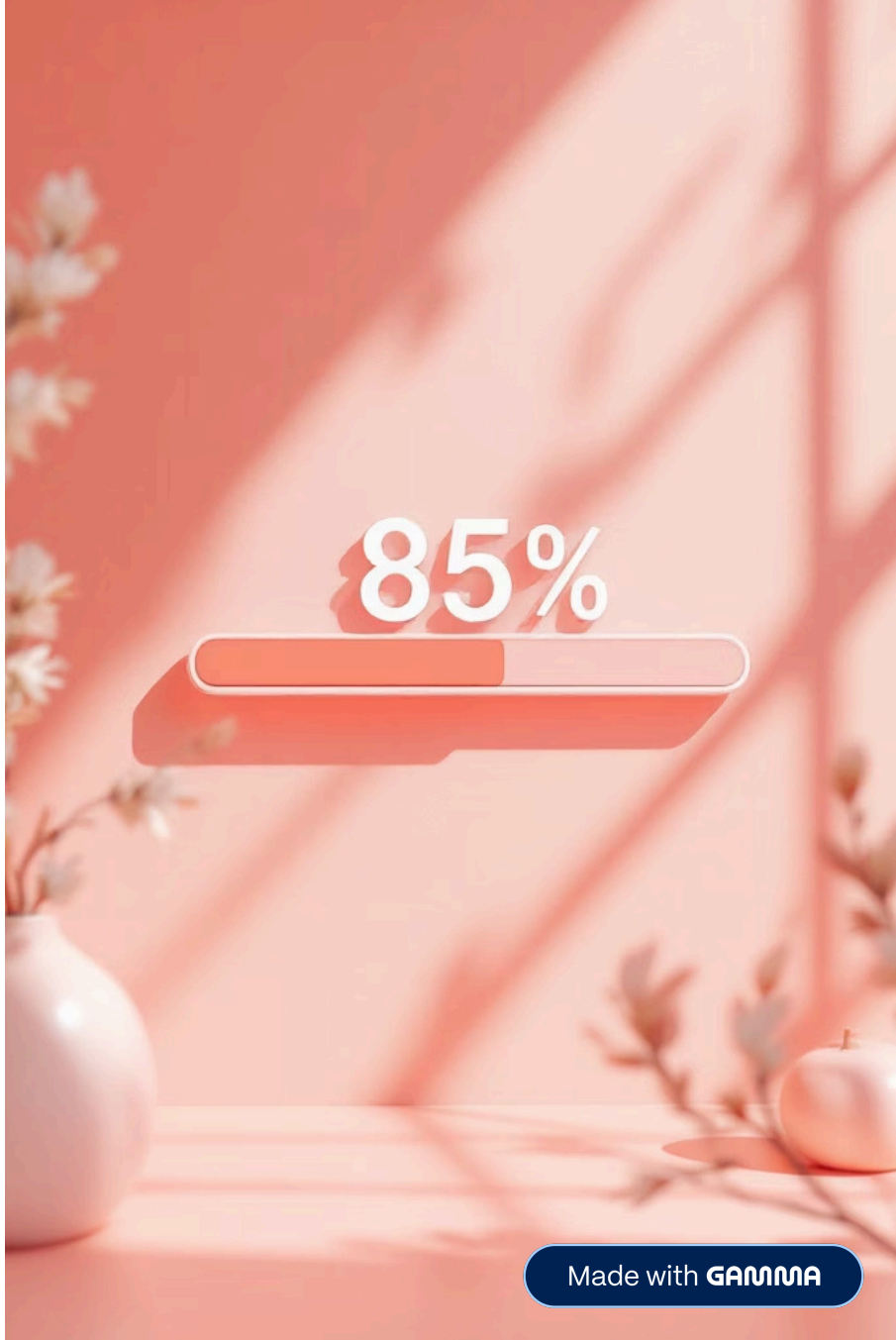
1. Helped Zeenat in recording datasets
2. Implementation of HW on veroboard with Noor

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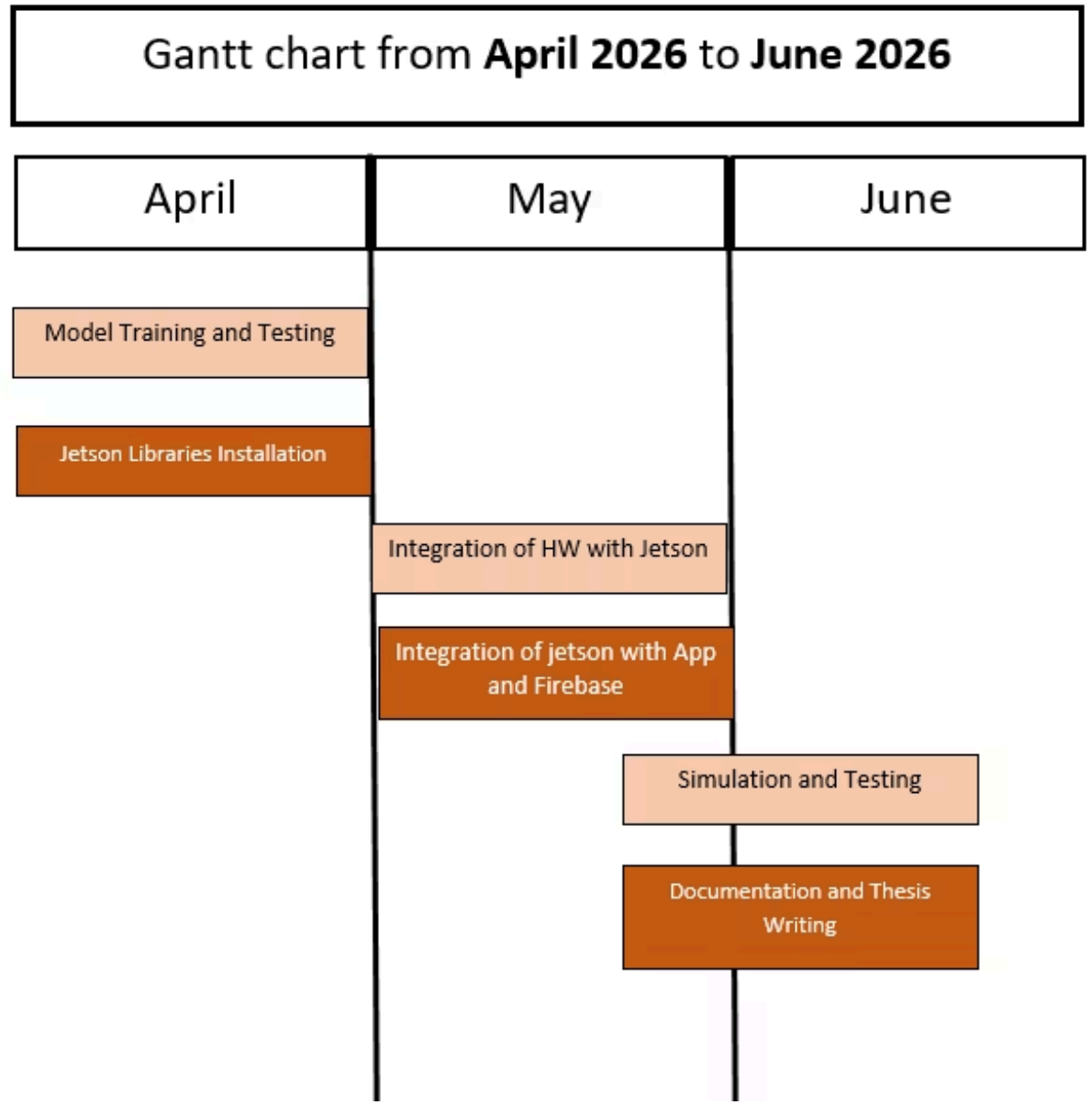
Current Project Progress

Done 85%:

- Dataset is created successfully.
- Tracking till software is complete.
- Model training is complete.
- Sensors implementation is complete.
- App is developed, alerts are generating.



Future Work (Gantt Chart)



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Conclusion

The assistance robot is in development phase as you know from our today's progress presentation.

The development of this project is almost done with too much dedication and hardwork.

Development in Progress

